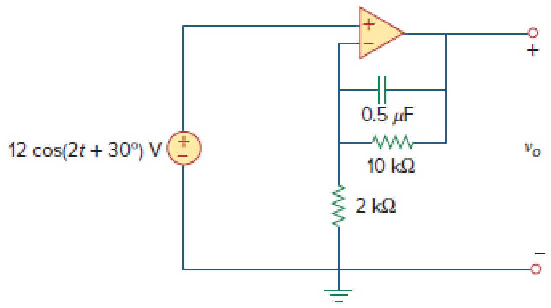
Chapter 10, Problem 71P

Problem

Find *vo* in the op amp circuit of Fig. 10.114.

Figure 10.114:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.114 in the textbook.

From the Figure 10.114, the angular frequency is, $\omega = 2 \text{ rad/s}$.

Convert the source voltage into phasor form.

$$12\cos(2t + 30°) = 12\angle 30° \text{ V}$$

Calculate the impedance of the capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)(0.5\times 10^{-6})} \\ &= -j10^6\ \Omega \end{aligned}$$

Step 2 of 4

Draw the circuit diagram in frequency domain.

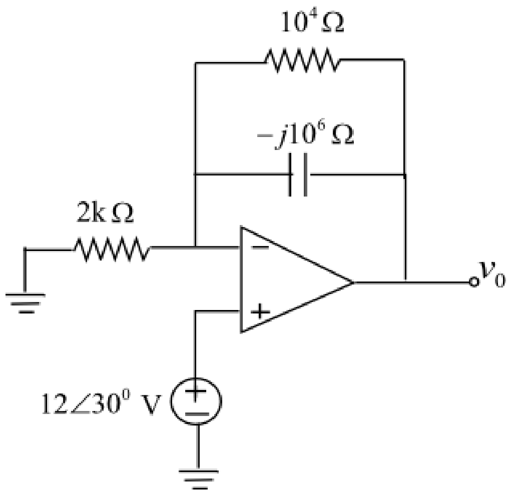


Figure 1

Step 3 of 4

From Figure 1, the feedback impedance is,

$$
\begin{aligned}
Z_f &= (10000) \| (-j10^6) \\
&= \frac{(10000)(-j10^6)}{(10000 - j10^6)} \\
&= 10000\angle -0.057^\circ\ \Omega
\end{aligned}
$$

The input impedance is,

$$Z_i = 2000\ \Omega$$

Since the circuit in Figure 1 is a non-inverting amplifier, the closed loop gain is,

$$
\begin{aligned}
\frac{\mathbf{V}_o}{\mathbf{V}_i} &= 1 + \frac{Z_f}{Z_i} \\
&= 1 + \frac{10000\angle -0.057^\circ}{2000} \\
&= 1 + 5\angle -0.057^\circ
\end{aligned}
$$

$$\frac{\mathbf{V}_o}{\mathbf{V}_i} = 6\angle -0.047^\circ \text{ ...... (1)}$$

Comments (1)

Anonymous

why is 1 + ?

Step 4 of 4

From Figure 1, the input voltage is,

$$V_i = 12\angle 30^\circ \text{ V}$$

From equation (1),

$$
\begin{aligned}
\mathbf{V}_o &= (6\angle -0.047^\circ)\mathbf{V}_i \\
&= (6\angle -0.047^\circ)(12\angle 30^\circ) \\
&= 72\angle 29.95^\circ \text{ V}
\end{aligned}
$$

In the time domain, the value of $v_0(t)$ is $v_0(t) = 72\cos(2t + 29.95^\circ)\text{ V}$.

Therefore, the output voltage, $v_0(t)$ of the op-amp is $\boxed{72\cos(2t + 29.95^\circ)\text{ V}}$.